Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Strain Gauge Sensor for Data Collection

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Introduction

- The shores of the Great Lakes are prone to bluff failure (Fig. 1). They can cause loss of lives and damage to local habitats and infrastructure, but the consequences can be lessened by identifying early warning signs.

Fig. 1: The red circle in this photo of bluff failure at Port Washington shows groundwater seeping out along the failure surface. This seepage might have caused this failure.

Objectives

- The goal of our research is to create a monitoring device that can accurately monitor mass movement so we can try to predict bluff failure.
- In this specific project, we work on gaining a better understanding of what the data collected by the strain gauge sensor signifies.
- To achieve this, we investigated real-life situations that can create slope failure and designed experiments based on these (Fig. 2).
- Additionally, we conducted experiments and calibrations to investigate the patterns observed in our data.

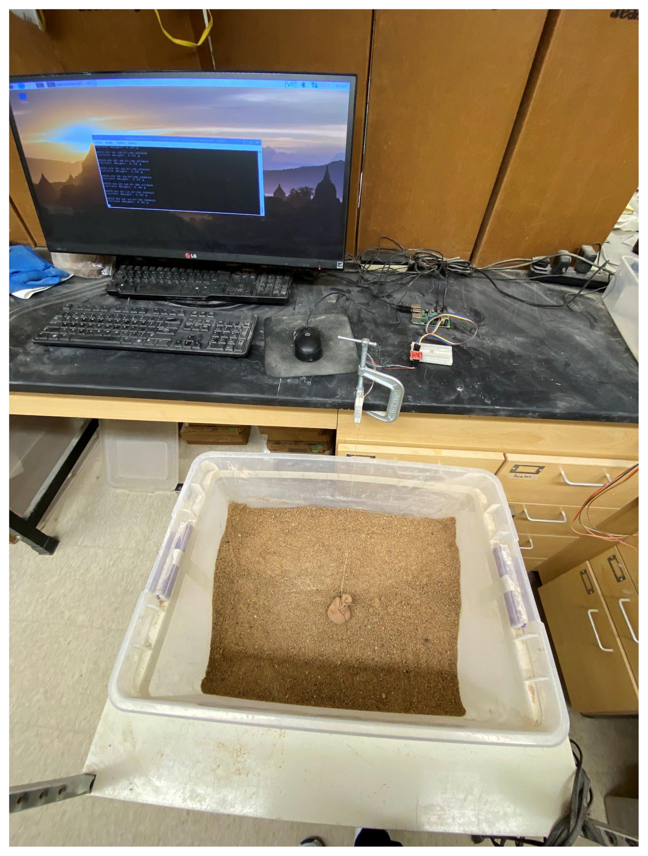

Fig. 2: Example of physical modeling setup.

Anchor Position Experiments

- From previous experiments done in our lab, we observed positive and negative trends in the plotted strain values.
- To investigate this, we conducted a set of experiments that explore the position of the buried anchor with the pattern of the plotted strain values (Fig. 3a & 3b).
  - 30 kg of total sand; 20 kg of 0.5 mm sand grains and 10kg of 0.25 sand grains.
  - Both layers were pre-saturated with DI water using a ratio of 100 cc/kg for the bottom layer and 80 cc/kg for the top layer to create a slope of 45°.
  - 60cc of DI water was added for 10 seconds over 30-second intervals until the bluff failure occurred.
- It was observed that rather than the depth, the angle created by the string and the z-axis that played a role in the trends observed in our data.
- A smaller angle leads to a negative trend, while a larger angle leads to a positive trend (Fig. 3c & 3d).

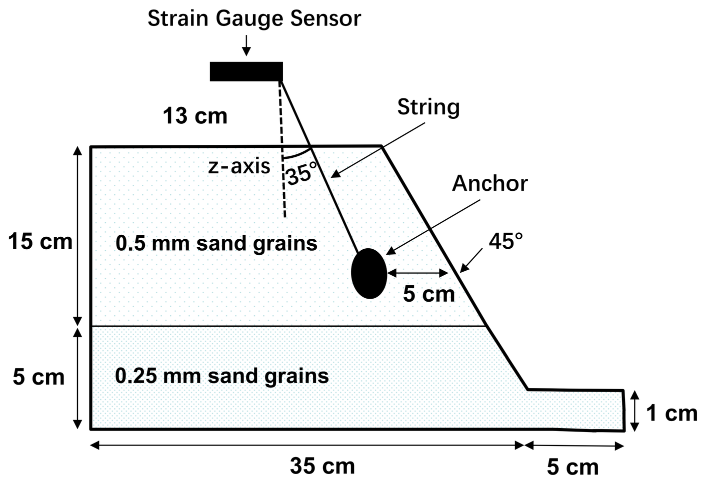

Fig. 3a: Physical modeling setup with 5 cm anchor burial depth and 35° angle.

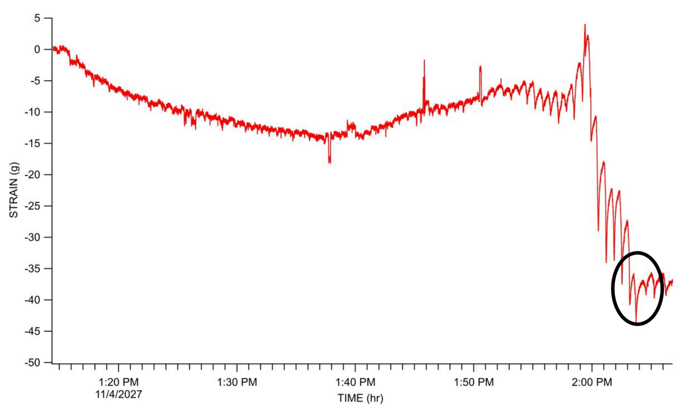

Fig. 3c: The black circle indicates failure.

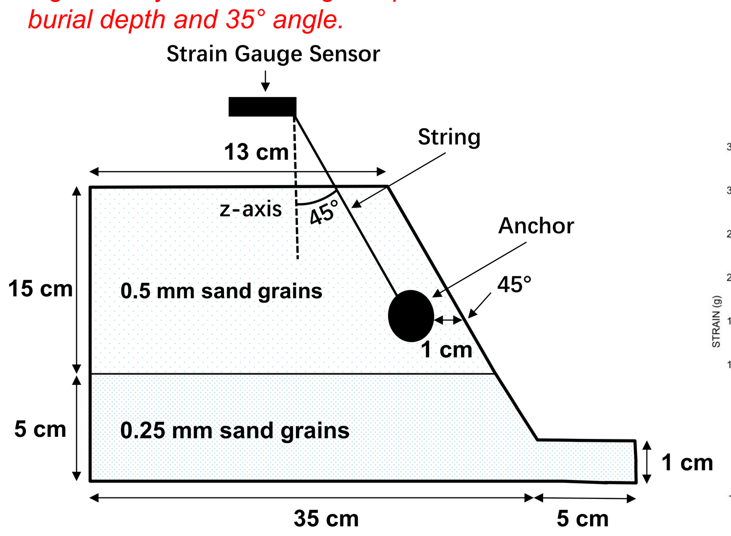

Fig. 3b: Physical modeling setup with 1 cm anchor burial depth and 45° angle.

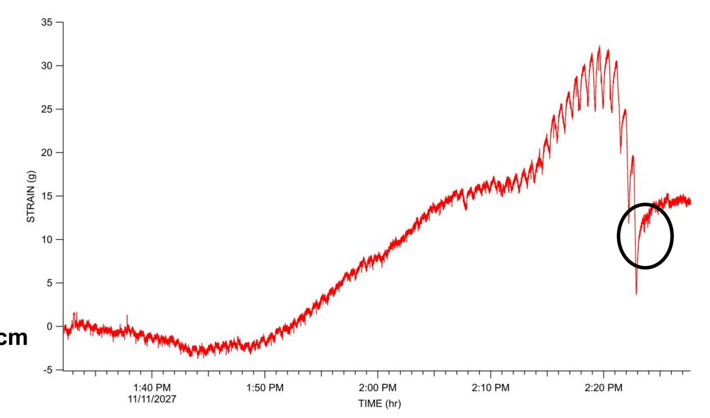

Fig. 3d: The black circle indicates failure.

Sensor Calibrations

- •We calibrated the strain gauge sensor to better understand the data.

Angle Calibrations

- The angle calibrations were done using a pulley to adjust a 400g mass attached to the sensor at a specific angle and changing the angle by 10° at every 10-minute intervals (Fig. 4a).
- Our observations are consistent with the calculated values for a range of angles (Fig. 4b & 4c).

Vertical Calibrations

- The vertical calibrations were done by increasing the mass by 50-gram increments for one experiment (Fig. 4e), and 5-gram increments in another (Fig. 4f).
- At 0°, there is almost a one-to-one relationship (Fig. 4e & 4f) between the mass added and the value recorded by our device.

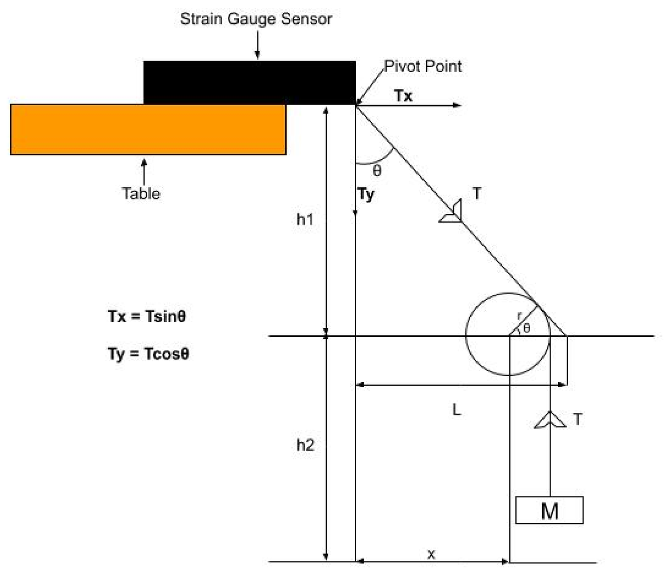

Fig. 4a: Diagram of setup used to conduct the angle calibration experiments.

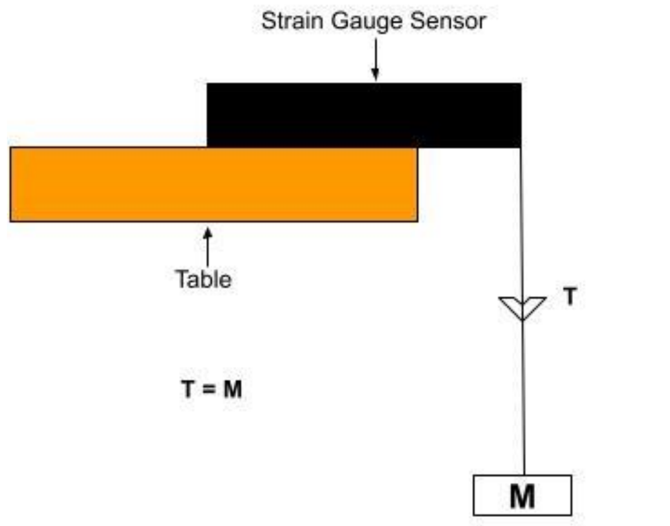

Fig. 4d: Diagram of setup used to conduct the vertical calibration experiments.

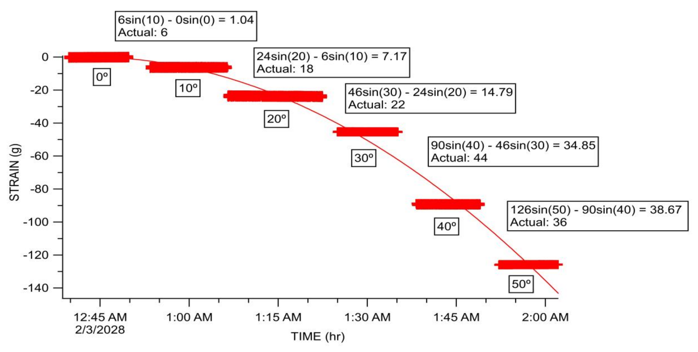

Fig. 4b: Results of angle calibration. Calculations made using $M_1 sin\theta_1 - M_2 sin\theta_2$.

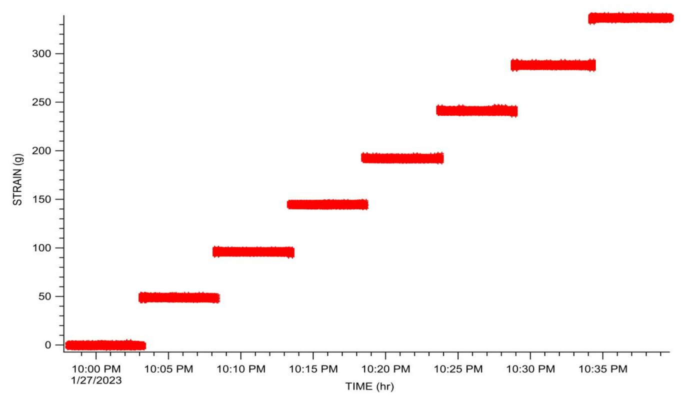

Fig. 4e: Results of vertical calibration with 50-gram increments.

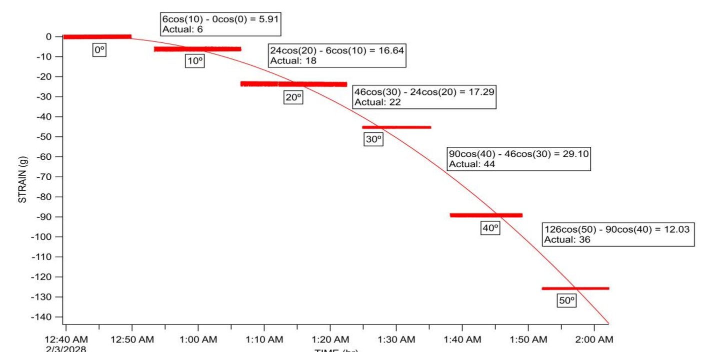

Fig. 4c: Results of angle calibration. Calculations made using $M_1 cos\theta_1 - M_2 cos\theta_2$.

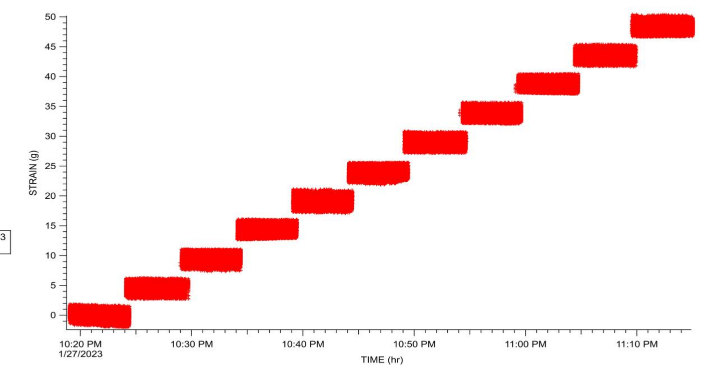

Fig. 4f: Results of vertical calibration with 5-gram increments.

Power Signal-to-Noise Experiments

- We investigated the power signal-to-noise ratio.
  - The dry setup (Fig. 5a) contained 30 kg of unsorted sand at a slope of 30°.
  - The moist setup was the same as the one presented in Fig. 5b on the anchor depth experiments.
  - In both cases, we left the setup untouched for 30 minutes and collected data (Fig. 5c & 5d).
- The dry setup displayed larger noise values but a smaller overall change in strain than the moist setup.

Dry

Fig. 5a: Physical modeling setup.

Moist

Fig. 5b: Physical modeling setup.

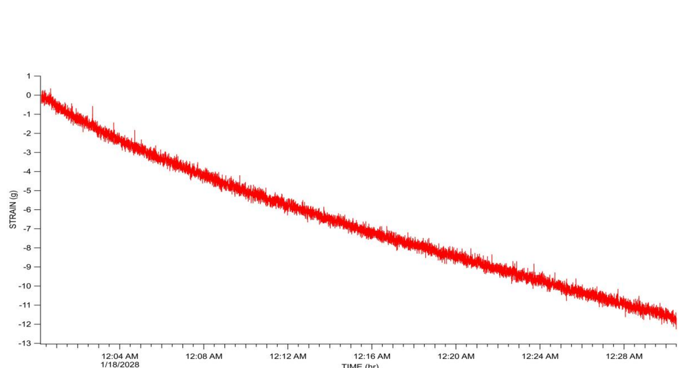

Fig. 5c: Results of experiments with a dry setup.

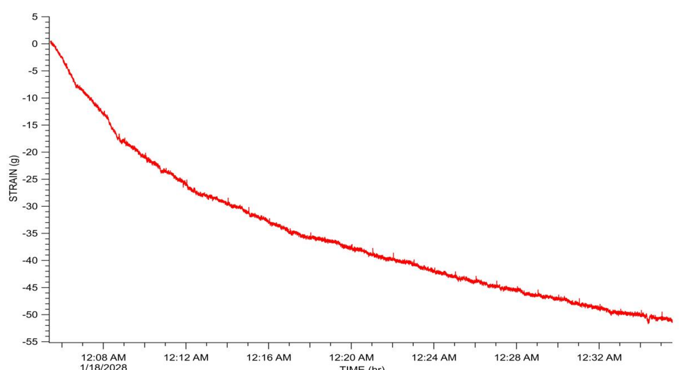

Fig. 5d: Results of experiments with a moist setup.

Water Volume Experiments

- Analyzing the plotted data led to the question of what the bumps in the strain values represent. It was hypothesized that these bumps were due to the volume of water added.
- We designed experiments to explore the relationship between the volume of water added and the bumps in the strain values.
  - The setup used was the same as the one presented in Fig. 3b on the anchor depth experiments.
  - 50cc of water was added for one experiment (Fig. 6a) and 200cc for another (Fig. 6b). In both, water was added for 20 seconds over 30-second intervals.
- In the results, we see larger bumps associated with a greater volume of water, while smaller bumps are associated with a smaller volume of water.
- Given that there was a difference in the data, it is possible that water volume impacts the data patterns.

50 cc

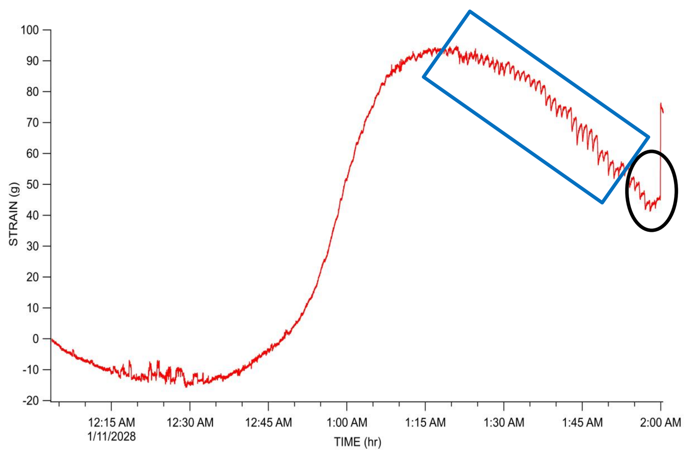

Fig. 6a: The black circle indicates failure. The blue rectangle shows the bumps observed in the data.

200 cc

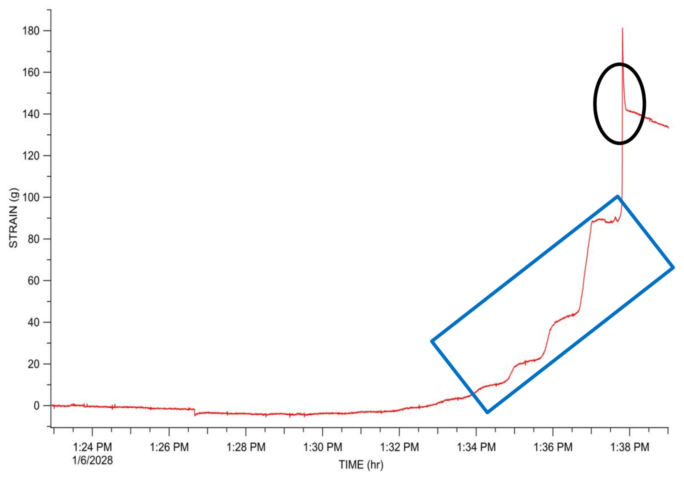

Fig. 6b: The black circle indicates failure. The blue rectangle shows the bumps observed in the data.

Water Addition Experiments

- The previous experiments raised the question of how the water addition rate impacts the data pattern.
- To test this, we designed experiments where the time interval between consequent water addition events was logarithmically spaced.
  - The setup used was the same as the one presented in Fig. 3b on the anchor depth experiments.
  - 200 cc of water was added at the time intervals of: 10, 16, 25, 40, and 63 seconds.
- The results show that the bumps in the data might be influenced by the water addition rate (Fig. 7).

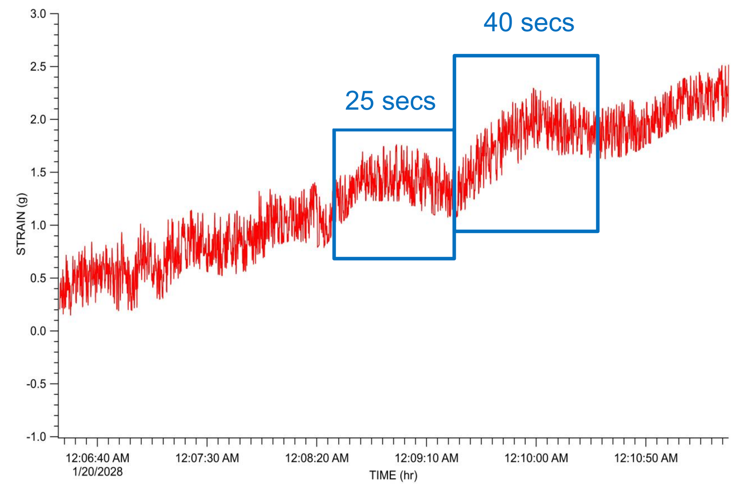

Fig. 7: The blue rectangles indicate changes in the strain, possibly influenced by specific time intervals between water additions.

Discussion

- These experiments allowed us to better interpret data from the strain gauge sensor under different experimental settings and conditions.
- This will help us interpret the data from our physical modeling experiments and ultimately help us predict slope failure in real life.

Going Forward

- We want to collect field data to compare with the data collected in the lab.
  - A team member is currently working on it.
- We want to incorporate another strain gauge sensor to simultaneously monitor strain at different locations.
  - Our Computer Science team members are working on the hardware and software.
- We want to further analyze the collected data.
  - A team member is currently learning different data analysis methods.
- We want to collect more experimental data (Fig. 8)
  - Changing the layer thickness and grain size
  - Incorporating liquid nitrogen for simulating freeze-thaw condition

Fig. 8: Example of simulated slope failure event.

Acknowledgements

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